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MSc DSA 055

Cricsheet T20I data

Llama 3.2-3B + QLoRA

# Motivation & Significance

## Why T20 score prediction matters

- Broadcast graphics and live commentary
- Fantasy and betting market projections
- Tactical planning for batting and bowling units
- Short format, high variance, rapidly changing innings state

TV

Stats

Market

Teams

## Core opportunity

Conventional models usually consume a flat numeric vector:

overs=12, runs=95, wickets=3, par=165

But analysts reason in context-rich language:

*"95 for 3 after 12 overs, with two strong batters still available, at a 165-par venue."*

**Research premise:** LLMs can exploit this semantic representation after domain-specific fine-tuning.

# Research Objectives & Scope

|            |                                                                        |            |                                                                    |
|------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------|
| <b>RO1</b> | Construct T20 snapshot dataset with engineered features including RBSS | <b>RO2</b> | Design natural-language prompt schema and prompt-completion corpus |
| <b>RO3</b> | Establish zero-shot Llama 3.2-3B baseline                              | <b>RO4</b> | Fine-tune Llama 3.2-3B using QLoRA SFT                             |
| <b>RO5</b> | Compare minimum-validation-loss and final checkpoints                  | <b>RO6</b> | Evaluate GPT-4.1-nano zero-shot as frontier reference              |

## In scope

- International men's T20 matches
- First-innings final score prediction
- Two-over in-game snapshots
- Parameter-efficient fine-tuning

## Out of scope

- Domestic leagues such as IPL, BBL, PSL
- Second-innings chase probability
- Full-parameter fine-tuning
- Multi-seed training analysis

# Literature Review (1): Themes & Trajectory

## Statistical cricket models

DLS/Stern resource framing  
Perera–Swartz Markov chains

*Strong interpretability, hand-crafted assumptions*

## Machine learning cricket models

Iyer & Sharda neural nets  
Kampakis & Thomas ensembles  
Bailey & Clarke wickets-in-hand

*Numeric vectors dominate*

## LLM foundations

Transformer, GPT, Llama  
LoRA and QLoRA

*Affordable domain adaptation becomes feasible*

## LLMs for structured prediction

TabLLM, LLM time-series forecasting  
Football outcome prediction

*Promising, but not yet T20 in-game scoring*

**Trajectory:** hand-crafted resources → numeric ML → language-model structured prediction.

## Literature Review (2): Research Gap & Contribution Map

| Study / stream                 | T20 prompt | RBSS-like | Small FT vs frontier | PEFT checkpoint |
|--------------------------------|------------|-----------|----------------------|-----------------|
| Perera–Swartz / Markov cricket | No         | No        | No                   | No              |
| Kampakis / ML cricket          | No         | No        | No                   | No              |
| Hegselmann / TabLLM            | No         | No        | No                   | No              |
| Huckerby / LLM football        | No         | No        | No                   | No              |
| PaleGreen <b>This thesis</b>   | Yes        | Yes       | Yes                  | Yes             |

### Synthesised gap

No prior work combines a natural-language T20 match-state prompt, RBSS-style batting-depth encoding, QLoRA fine-tuning of a compact open-source LLM, and a head-to-head benchmark against a frontier zero-shot model.

# Methodology Overview



## Data hygiene

No-result and incomplete first innings removed.

## Leakage control

Match-level temporal split before evaluation.

## Artefacts

HF datasets, HF adapter, W&B logs, Modal service.

# Feature Engineering & RBSS

## Snapshot construction

- First innings divided into ten 2-over sections
- One prompt-completion pair per section
- Feature groups: state, recent momentum, phase, venue, RBSS
- Venue par score: 52nd percentile of historical first-innings totals

## RBSS definition

$$\text{RBSS} = \sum_{p \in \mathcal{R}} w(\text{class}(p))$$

$\mathcal{R}$  = undischarged players at the end of the section.

| Player class      | Weight |
|-------------------|--------|
| Batter            | 1.00   |
| All-rounder       | 0.80   |
| Utility player    | 0.65   |
| Insufficient data | 0.45   |
| Bowler            | 0.30   |

## Worked example

$$1(1.00) + 2(0.80) + 1(0.65) + 3(0.30) = 4.15$$

**Why it matters:** identical scorelines can have different remaining batting quality.

# Prompt Schema & Model Architecture

## Prompt skeleton

```
Match type: T20
Venue: {venue}
Venue par score: {par}
Batting team: {team}

Overs completed: {overs}
Balls remaining: {balls}
Runs scored: {runs}
Wickets lost: {wickets}
Current run rate: {rate}

Runs in last 2 overs: {recent}
Dot balls so far: {dots}
Fours hit: {fours} ; Sixes hit: {sixes}
Powerplay completed: {Yes/No}
Remaining Batting Strength Score: {RBSS}
Final 1st innings score:
```

max prompt length: 148 tokens

max seq length: 256

## QLoRA fine-tuning recipe

|              |                                        |
|--------------|----------------------------------------|
| Base model   | Llama 3.2-3B decoder-only              |
| Quantisation | 4-bit NF4 + double quantisation        |
| LoRA         | $r = 16$ , $\alpha = 32$ , dropout 0.1 |
| Targets      | q_proj, k_proj, v_proj, o_proj         |
| Training     | 1 epoch, lr $2 \times 10^{-4}$ cosine  |
| Hardware     | Kaggle NVIDIA T4, fp16, batch eff. 8   |

Frozen 4-bit  
Llama weights

Trainable  
LoRA adapters

Integer  
score output



# Data, Splits & Tools

## Strict temporal split

| Split      | Date range               | Matches | Rows   |
|------------|--------------------------|---------|--------|
| Train      | Up to 2024-12-31         | 2,408   | 24,080 |
| Validation | 2025-01-01 to 2025-06-30 | 237     | 2,370  |
| Test       | 2025-07-01 onwards       | 250     | 2,500  |

- Match-level chronological partitioning
- Simulates future deployment conditions
- Reduces risk of temporal leakage

## Reproducibility stack

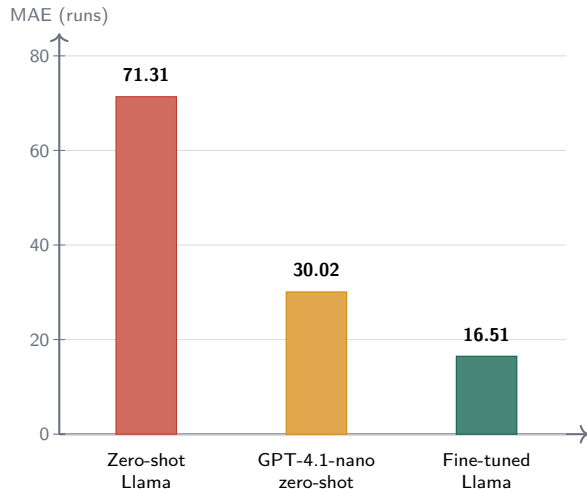
Pydantic v2, SQLite, fuzzywuzzy, Hugging Face Datasets, PEFT, TRL, BitsAndBytes, W&B, LiteLLM, Modal, Gradio, React/Vite, Azure Static Web Apps.

W&B run: v64wpa10

HF adapter: 00be3f3f

Dataset: 28,950 rows

# Results: Fine-Tuned 3B Beats Zero-Shot Frontier



## Cross-condition metrics

| Model           | MSE   | $R^2$   |
|-----------------|-------|---------|
| Llama ZS        | 7,463 | -317.4% |
| GPT-4.1-nano ZS | 1,538 | 14.0%   |
| Llama FT        | 543   | 68.3%   |

vs Llama ZS: 76.8% lower MAE

vs GPT-4.1: 45.0% lower MAE

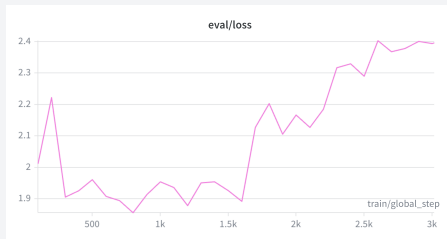
$R^2$  gain: +54.3 pp vs GPT-4.1

# Discussion: Why Fine-Tuning Wins

## Interpretation

- General pre-training sees cricket prose, not the conditional mapping  $P(\text{final} \mid \text{state})$ .
- SFT supplies 24,080 domain examples of that mapping.
- GPT-4.1-nano tends to under-project early states: e.g. 14/1 after 2 overs at par 160 → predicted 89, actual 160.
- RBSS and venue par score enrich the prompt beyond raw score and wickets.

## Checkpoint selection



Minimum validation loss: **1.856 at step 800**

| Checkpoint | N   | MAE          | $R^2$        |
|------------|-----|--------------|--------------|
| Step 800   | 500 | <b>16.51</b> | <b>68.3%</b> |
| Final      | 500 | 16.54        | 68.1%        |

# Limitations

## Scope limitations

- International men's T20 only
- Domestic leagues excluded
- First innings only
- Women's matches not modelled in this study

## Evaluation boundaries

- GPT-4.1-nano evaluated at  $n = 200$  due to API cost
- No classical XGBoost or linear-regression baseline

## Methodological limitations

- Single fine-tuning run; no multi-seed variance
- 3B base model only; larger open-source models untested
- Validation loss computed on 50-row subsample
- Greedy point prediction only; no uncertainty intervals
- Regex fallback-to-zero inflates zero-shot Llama error

# Conclusions & Future Work

## Conclusions

- 1 Domain-specific fine-tuning transforms zero-shot Llama from unusable to competitive.
- 2 A compact fine-tuned open-source model beats a frontier zero-shot model on this structured numerical task.
- 3 Minimum-validation-loss checkpoint selection provides a small but consistent generalisation benefit.

**Core claim:** small, fine-tuned, domain-specific LLMs can be practical sports forecasting tools.

## Future work

- Extend to IPL, BBL, PSL and other domestic leagues
- Evaluate 7B, 8B and 70B open-source models
- Build second-innings chase prediction
- Add uncertainty via sampled completions
- Run feature ablations, especially RBSS and venue par
- Integrate live ball-by-ball feeds

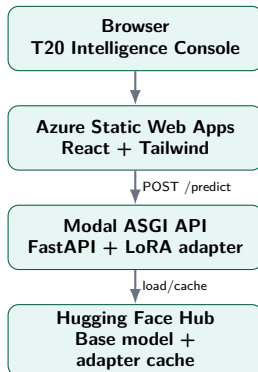
# Practical Implications

## For cricket analytics teams

- Prioritise domain data curation and PEFT over frontier API prompting
- Host small fine-tuned models on controlled infrastructure
- Use point predictions for graphics and simulation, not high-stakes settlement

## Deployment validation

- Modal GPU backend: about 30 s cold, about 2 s warm
- Gradio prototype on Hugging Face Spaces
- React/Vite dashboard on Azure Static Web Apps



<https://mango-ocean-00260de00.7.azurestaticapps.net/>

# References (Selected)

- ❶ Vaswani et al. (2017). *Attention Is All You Need*.
- ❷ Brown et al. (2020). *Language Models are Few-Shot Learners*.
- ❸ Touvron et al. (2023). *LLaMA: Open and Efficient Foundation Language Models*.
- ❹ Hu et al. (2022). *LoRA: Low-Rank Adaptation of Large Language Models*.
- ❺ Dettmers et al. (2023). *QLoRA: Efficient Finetuning of Quantized LLMs*.
- ❻ Heggelmann et al. (2023). *TabLLM: Few-shot Classification of Tabular Data with LLMs*.
- ❼ Gruver et al. (2023). *Large Language Models are Zero-Shot Time Series Forecasters*.
- ❽ Huckerby et al. (2024). *Leveraging LLMs for Football Match Outcome Prediction*.
- ❾ Duckworth and Lewis (1998). *A Fair Method for Resetting Targets*.
- ❿ Bailey and Clarke (2004). *Predicting Match Outcome in ODI Cricket*.
- ⓫ Cricsheet (2024). *A Cricketing Archive*.

Full bibliography appears in the thesis reference list.

## Thank you. Questions?